

CEM: more arguments

Topics in Mind and Language: Conditionals

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1. Unfinished logic-related business

(i) Given our background logic, CEM entails And-to-if

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| 1. | A | premise |
| 2. | B | premise |
| 3. | $A > \neg B$ | assumption |
| 4. | $\neg B$ | 1,3, MP |
| 5. | $\neg(A > \neg B)$ | 2–4, reductio |
| 6. | $(A > B) \vee (A > \neg B)$ | CEM |
| 7. | $A > B$ | 5,6, disjunctive syllogism |

(ii) Given our background logic, the combination of And-to-if with Antecedent Strengthening entails Or-to-if (i.e. the material conditional analysis)

(here $\top$ represents some arbitrary tautology, e.g. $B \vee \neg B$)

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|----|---------------------------------------|---|
| 1. | $A \vee B$ | premise |
| 2. | $\top$ | classical propositional logic |
| 3. | $\top > (A \vee B)$ | 1, 2, And-to-if |
| 4. | $(\neg A \wedge \top) > (A \vee B)$ | 3, Antecedent Strengthening |
| 5. | $\neg A > (A \vee B)$ | 4, substitution of equivalent antecedents |
| 6. | $\neg A > \neg A$ | ID |
| 7. | $\neg A > (\neg A \wedge (A \vee B))$ | 5, 6, CC |
| 8. | $\neg A > B$ | 7, consequent weakening |

(iii) CEM entails the validity of the following principle:

Disjunction Distribution: $A > (B \vee C) \vdash (A > B) \vee (A > C)$

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| 1. | $A > (B \vee C)$ | premise |
| 2. | $(A > B) \vee (A > \neg B)$ | CEM |
| 3. | $A > \neg B$ | assumption |
| 4. | $A > ((B \vee C) \wedge \neg B)$ | 1, 3, CC |
| 5. | $A > C$ | 4, consequent weakening |
| 6. | $(A > B) \vee (A > C)$ | 2–5, proof by cases |

• Conversely, the validity of Disjunction Distribution entails that of CEM:

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| 1. | $A > A$ | ID |
| 2. | $A > (B \vee \neg B)$ | 1, consequent weakening |
| 3. | $(A > B) \vee (A > \neg B)$ | 2, disjunction distribution |

(iv) Given the closeness analysis, we can get CEM to come out valid by imposing two further constraints on the closeness relation (in addition to transitivity, connectedness, and weak centering):

Antisymmetry If $u \leq_w v$ and $v \leq_w u$, then $u=v$.

The Limit Assumption If S is a nonempty set of worlds and w is any world, then there is some $u \in S$ such that for every $v \in S$, $u \leq_w v$.

- Indeed, the validity of CEM *entails* that Antisymmetry holds, at least in the case where u and v are both accessible from w . For let p_u be any proposition that is true at u and no other possible world, and let p_v be a proposition true at v and no other possible world. Then given the Closeness Analysis, $(p_u \vee p_v) > p_u$ is true at any world w just in case either (i) neither u nor v is accessible from w , or (ii) v is accessible from w and u is not, or (iii) both u and v are accessible from w and it's false that $v \leq_w u$, or (iv) $u=v$.
- Given Disjunction Distribution and ID, $((p_u \vee p_v) > p_u) \vee ((p_u \vee p_v) > p_v)$ is true at every world w , for any worlds u and v . So if u and v are both accessible from w , this means that one of the following conditions obtain: either $u=v$, or $v \leq_w u$ is false, or $u \leq_w v$ is false. In other words: if $u \leq_w v$ and $v \leq_w u$, then $u=v$.
- Something similar is true for the Limit Assumption.

2. The duality argument against CEM (for counterfactuals)

Duality: 'If A , it might be the case that B ' is equivalent to 'It is not the case that: if A , it would be the case that not- B '.

- Or to put it another way: 'Not (if A , might B)' is equivalent to 'If A , would not- B '.
- The left-to-right entailment here is pretty uncontroversial; what's controversial is the right-to-left direction, i.e. the claim that 'If A , would not- B ' and 'If A , might B ' are *inconsistent*.

If we assume both Duality and CEM, we get the crazy result that 'If A , might B ' *entails* 'If A , would B ':

1.	If A , might B	Premise
2.	Not (If A , would not- B)	Duality
3.	$(\text{If } A, \text{ would } B) \vee (\text{If } A, \text{ would not-}B)$	CEM
4.	If A , would B	2, 3, disjunctive syllogism

So either Duality or CEM has to go.

Why believe Duality? Basic datum: conjunctions of the form 'If A , would not- B , and if A , might B ', or 'If A , might not- B and if A , would B ' sound terrible.

- First worry about this argument: by parity of reasoning it leads to the conclusion that future-directed ‘might’ conditionals are dual to ‘will’ conditionals: e.g. ‘If I run I might be elected’ is inconsistent with ‘If I run I will be defeated’.
- Second worry about this argument: ‘It might be raining and it isn’t raining’ also sounds bad. But it looks like a terrible idea to say that this conjunction is *inconsistent*, since this would make the argument from ‘It might be raining’ to ‘It is raining’ valid. Are we sure we’re not mistaking this other form of badness for the badness of inconsistency?

Challenge: find an alternative account of the meaning of ‘If A, might B’ that can explain the badness.

Wide scope option: ‘If A, might B’ is equivalent to ‘It might be that (if A, it would be that B)’

Narrow scope option: ‘If A, might B’ is equivalent to ‘If A, it would be that (it might be that B)’.

Lewis’s objection (*Counterfactuals* p. 80): ‘If I had looked in my pocket, I might have found a penny’ is false in the case where I don’t have a penny in my pocket, but on these analyses it’s true so long as it’s consistent with what I know that there is just a penny in my pocket and also consistent with what I know that there is just a dime there.

- This assumes that ‘Might P’ is equivalent to ‘P is consistent with what I know’. (Or maybe ‘P is consistent with what I know’.)
- But this may be too simple. Consider: ‘I don’t know whether this coin might land heads, because I don’t know whether it’s a double headed coin’. Sometimes, the set of worlds relevant to the truth of a ‘might’ claim seems to be narrower than the set of all worlds consistent with what we know—in this case, ‘might P’ perhaps means something more like ‘P is consistent with what I know and with the truth about what kind of coin it is’. Maybe in Lewis’s case, the operative ‘might’ is one on which ‘might P’ is equivalent to ‘P is consistent with what I know and with the truth about how many coins are in my pocket’?
 - However if we want to go for a version of the wide scope option using this kind of ‘might’, we’d like a story about why ‘might’ gets constrained in this way.

3. An argument for CEM: from quantified conditionals

Higginbotham (****) notes the apparent equivalence of (1a) and (1b), and raises this as a puzzle for semanticists:

- (1) a. Every student will fail if he goofs off
- b. No student will pass if he goofs off

We get the same apparent of equivalence when we consider counterfactuals or past-tense indicatives:

- (2) a. Every student would have failed if he had goofed off
b. No student would have passed if he had goofed off
- (3) a. Every student failed if he goofed off
b. No student passed if he goofed off

Let's focus for now on (2a) and (2b). And let's assume they have the logical forms they seem to have, ignoring the option of securing the equivalences by messing with logical form.

For the argument from (2a) to (2b) to be strictly valid, ' x would have failed if he had goofed off, and x would have passed if he had goofed off' would have to be inconsistent. None of the views we have considered says that this is inconsistent, since none of them says that the schema CNC ('Conditional non-contradiction') is valid:

$$\text{CNC} \quad \neg((A > B) \wedge (A > \neg B))$$

(Note that no view on which ID and Weakening the Consequent are both valid can validate CNC, since the instance $(A \wedge \neg A) > (A \wedge \neg A)$ of ID implies both $(A \wedge \neg A) > A$ and $(A \wedge \neg A) > \neg A$ by Weakening the Consequent.) However, we have a resource for explaining the seeming goodness of the inference from (2a) to (2b), namely the *presupposition of non-vacuity*. According to this idea, an utterance of $A > B$ carries the presupposition that some A -worlds are accessible (from the actual world), which entails that corresponding instance of CNC is true.

The direction we need to focus on, then, is the argument from (2b) back to (2a). This is strictly valid just in case 'Not (x would have passed if he had goofed off), and not (x would have failed if he had goofed off)' is inconsistent. Von Fintel and Iatridou (2002) point out that if CEM is valid, this is inconsistent, and (2b) does imply (2a). If CEM *isn't* valid, it is hard to see why (2b) would even seem to imply (2a): why would we ignore the possibility that some of the students are neither such that they would have passed if they had goofed off, nor such that they would have failed if they had goofed off?

4. A possible response: CEM as a mere presupposition

Von Fintel's suggestion: CEM has the same status as CNC. 'If A , B ' *presupposes* 'Either if A , B or if A , not- B '.

Combine this with a general claim about presupposition projection: if ' x is F ' presupposes ' x is G ', 'Every H is F ' and 'No H is F ' both presuppose 'Every H is G '.

Result: both (5a) and (5b) presuppose 'Every student either will fail if he goofs off, or will pass if he goofs off'.

So: (5b) is true *and its presupposition is satisfied*, (5a) is true. (And vice versa.)

This proposal also gives a response to some of the arguments about scope. But it doesn't seem to the arguments about chance and credence.

5. Against CEM: from the similarity account of closeness

Even if we are, like Lewis, pretty liberal about what counts as a comparative similarity relation, it's hard to see how any comparative similarity relation could be antisymmetric.

One could retreat from the view that closeness *is* a comparative similarity relation to the view that there is a comparative similarity relation such that whenever *u* is more similar to *w* than *v* is in that sense, *u* is closer to *w* than *v* is (but not vice versa)!

- Note however that if the Limit Assumption fails for the relevant comparative similarity relation, this sufficient condition means that it's also got to fail for closeness, which means we still can't have CEM. And as Lewis points out, the Limit Assumption fails for many comparative similarity relations.
- Also: the kind of reasoning that drives the chance- and credence-based arguments for CEM fits poorly with the comparative similarity vision.
 - Suppose we tossed a fair coin five times yesterday, starting at noon, and it landed Heads on tosses 3 and 5 and Tails on tosses 1, 2, and 4. Q: How likely was it, at noon, that it would land Heads five times, if it landed Heads at least four times?
 - The kind of reasoning we've been relying on suggests the answer '1/6', based on the fact that there are 6 equiprobable ways for it to land Heads at least four times, of which one involves it landing Heads five times. But if being more similar to actuality is sufficient for being closer to actuality, and the number of coin outcomes matters at all, it looks like the answer's going to be 1/64. So long as the coin doesn't *actually* land Heads every time (something whose chance is 1/64), some worlds where it lands Heads exactly 4 times will be more similar to actuality than any world where it lands Heads 5 times.
- Tentative conclusion: if you want to hold on to CEM, you had better forget about the idea that closeness has anything to do with similarity. We know more about similarity than we know about closeness—for example, even knowing how the coin landed, we just don't know whether all-Heads worlds are closer than four-Heads worlds (our credence is 1/6).

6. Against CEM: 'what could make it true'?

7. Background on vagueness and classical logic

We have been assuming classical logic, including the validity of the schema LEM:

LEM $A \vee \neg A$

If this is valid, then so its substitution instance

(H) Either Harry is bald, or Harry is not bald

Some philosophers think that (H) must be rejected in the case where Harry is border-line-bald. One of their basic arguments is a ‘what could make it true?’ argument: the thought is that there is nothing in our use of the word ‘bald’ that could make ‘Harry is bald’ true or false.

Danger! If you say that it’s *not* the case that (either Harry is bald or Harry is not bald), it seems to follow (by a rule that even those who reject some bits of classical logic in this context tend to like) that Harry is not bald and also not not bald. But this is a contradiction (which is again something that most people don’t want to be committed to, even in this context).

But many (most?) philosophers thinking about vagueness accept (H).

The *semantic indecision* theory of vagueness: there are many different properties that are ‘admissible interpretations’ of the word ‘bald’, some of which are had by Harry and some of which are lacked by Harry. But (H) is *supertrue* — all the propositions that are admissible (uniform) interpretations of (H) are true. So we can assert (H)

- Many philosophers in this ‘supervaluationist’ tradition (but not Lewis) add: truth for sentences = supertruth. So a disjunction can be true without having a true disjunct.
- Note that this is specific to ‘true’ when applied to *sentences*. There is no reason I can see to carry it over to ‘true’ as a predicate of *propositions*, or to ‘it is true that ...’ — we can say the same thing about ‘Either it is true that Harry is bald, or it is true that Harry is not bald’ that we said about H itself.

8. Back to CEM